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Dept: Computer Science and Engineering

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Subject: Computer Networks Lab

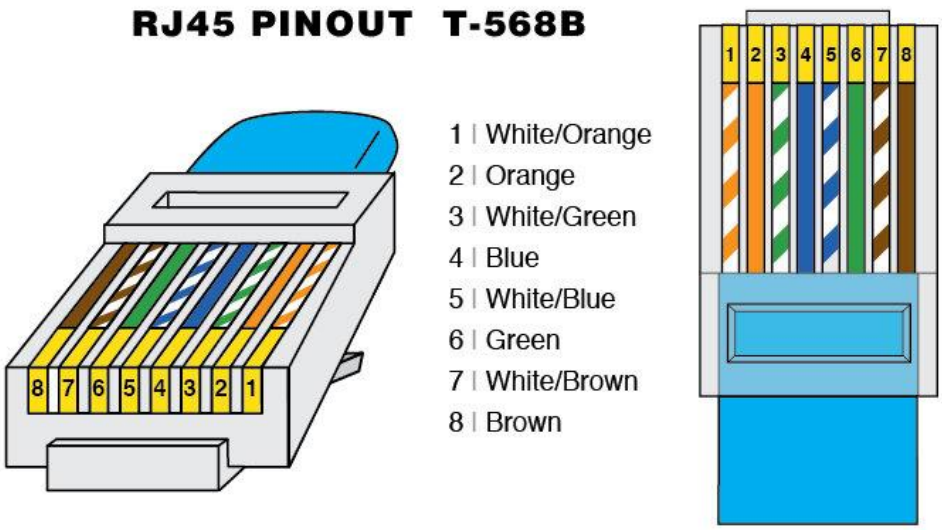
Code: PCC-CS692

PCA1 - Lab assignment 1

1) Cat 6 Cable & Colour Combinations

What is a Cat 6 Cable?

Category 6 (Cat 6) is a standardized twisted-pair Ethernet cable used for high-speed wired networks.



Key specifications:

- Supports speeds up to **1 Gbps** (Gigabit Ethernet) up to 100 meters
- Can support **10 Gbps** up to ~55 meters (Cat6a supports 10 Gbps up to 100 m)
- Bandwidth: **250 MHz**
- Used for LAN, routers, switches, PCs, CCTV, gaming, etc.
- Contains **4 twisted pairs = 8 copper wires**

Structure of Cat 6 Cable

Inside the cable:

- 8 wires arranged in 4 pairs
- Each pair is twisted to reduce interference (crosstalk)
- Often includes a plastic spline (separator) in the centre for performance

Four Twisted Pairs

Each pair consists of:

1. Solid colour wire
2. White wire with matching colour stripe

◈ Standard Colour Codes

There are **two official wiring standards**:

T568A Standard

Pin — Wire Colour

1 — White/Green
2 — Green
3 — White/Orange
4 — Blue
5 — White/Blue
6 — Orange
7 — White/Brown
8 — Brown

T568B Standard (Most Common)

Pin — Wire Colour

1 — White/Orange
2 — Orange
3 — White/Green
4 — Blue
5 — White/Blue
6 — Green
7 — White/Brown
8 — Brown

👉 In India and most networks, **T568B is commonly used**

◈ Straight-Through vs Crossover Cable

▯ Straight-Through Cable

Same standard on both ends (A-A or B-B)

Used for:

- PC ↔ Switch
- Router ↔ Switch
- PC ↔ Router

Most common: **B-B**

⛌ Crossover Cable

Different standards on each end (A-B)

Used for:

- PC ↔ PC (old networks)
- Switch ↔ Switch (without auto-MDI/MDIX)

Now rarely needed due to auto-sensing ports.

◈ RJ45 Connector Pinout Purpose

Pins used for data transmission in Gigabit Ethernet:

- All 8 wires are used
- Earlier Fast Ethernet used only 4 wires

◈ Why Twisting Matters

Twisting:

- Reduces electromagnetic interference
- Prevents signal loss
- Improves data reliability

Different pairs have different twist rates.

◈ Shielding Types (Optional Info)

Cat 6 cables may be:

- UTP — Unshielded Twisted Pair (most common)
- FTP — Foil Shielded
- STP — Shielded Twisted Pair

Shielding protects against electrical noise.

2) Different Types of Network Devices

1⃣ Hub

Features:

- Basic networking device connecting multiple computers
- Broadcasts data to all connected devices
- Operates at Physical Layer (Layer 1)

Advantages:

- Low cost
- Easy to install
- Suitable for very small networks

Disadvantages:

- High data collisions
- No data filtering
- Slow and outdated

2⃣ Switch

Features:

- Connects devices in a LAN
- Sends data only to the destination device
- Uses MAC address table
- Works at Layer 2

Advantages:

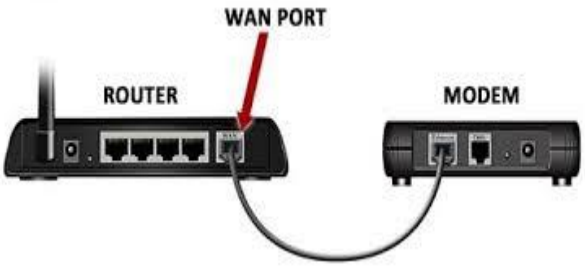
- Faster than hub

- Reduces collisions
- Better performance

Disadvantages:

- Higher cost than hub
- Cannot connect different networks alone

3⃣ Router



Features:

- Connects multiple networks
- Routes data using IP addresses
- Provides Internet access
- Works at Layer 3

Advantages:

- Connects LAN to Internet
- Provides security features
- Supports multiple devices

Disadvantages:

- Expensive
- Needs configuration
- Slight delay in routing

4⃣ Modem

Features:

- Converts digital ↔ analog signals
- Connects ISP to user network
- Provides Internet signal

Advantages:

- Essential for Internet access

- Enables broadband connectivity

Disadvantages:

- Limited functions
- Requires router for Wi-Fi

5⃣ Access Point (AP)

Features:

- Provides wireless network access
- Connects Wi-Fi devices to wired LAN
- Extends network coverage

Advantages:

- Supports wireless connectivity
- Increases mobility
- Expands coverage area

Disadvantages:

- Security risks if unsecured
- Signal interference possible
- Additional cost